

Personalized E-learning with Adaptive Quiz Difficulty and Mastery Progression

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1 INTRODUCTION

In the aftermath of COVID-19, educational institutions have widely adopted online learning [23, 50]. This shift is particularly beneficial for low- and middle-income countries where access to quality education is limited [51]. However, since one-size-fits-all approaches are rarely sufficient [16], there is an increasing need for personalized and adaptive applications in e-learning [25].

Such systems - known as Intelligent Tutoring Systems (ITS) [8], adaptive learning technologies [66], or adaptive educational systems (AES) [60], among other terms - employ user models that consider learners' knowledge, interests, and goals to mimic human-like tutoring. By guiding learners and dynamically adjusting instructional strategies, materials, and assessments [16, 46], they improve learning outcomes across educational levels [37, 62]. Additionally, they address usability challenges beyond traditional interface-focused methods [14] and enhance student interest [58], engagement [26], and satisfaction [68].

Despite the AI boom and many commercially successful applications [44], ITS still fall short of their initial promise due to three key gaps. First, they rely on simplistic methods and do not fully leverage AI capabilities [10]. Advances in natural language processing (NLP), such as large language models (LLMs) that have revolutionized education [27], have not been widely integrated into commercial applications, even though this is arguably feasible and scalable [48]. Second, the focus on STEM fields limits domain coverage, with minimal representation of the humanities and exclusion of niche domains lacking expert-defined knowledge representations [52, 61]. Finally, these systems often lack sufficient adaptivity and theoretical grounding. Commercial applications that do use advanced NLP focus mainly on automated assessment or content generation, without integrating pedagogically informed knowledge representations or advanced, learner-centered, interactivity and personalization [24, 25].

In this study, we propose a system that addresses these limitations by leveraging recent advances in NLP to automatically generate learning resources with broad domain coverage and advanced personalization features grounded in sound and well-established pedagogical principles. Recognizing the challenges higher-education students face when navigating extensive, complex material, our solution allows users to upload their own textbooks and interact with the content in small, digestible segments, and enables iterative self-assessment and progression at their own pace and desired level of understanding. By

harnessing the transformative power of LLMs, we integrate the functionalities of existing systems into a highly adaptive, general-purpose tool, reducing the need for costly development of specialized domain knowledge models across diverse and, often, under-represented, fields of study.

We anticipate that the proposed system will lead to improved learning outcomes and increased user satisfaction. Below we present related literature, the system and the modeling approach, and we propose a framework for its formal evaluation.

2 RELATED WORK

2.1 Pedagogical Frameworks in Adaptive Learning Systems

Learning styles—combinations of cognitive, affective, and physiological factors that reflect how learners prefer to process information—are one of the most common methods used in adaptive learning environments to tailor educational experiences [7, 64]. While these systems aim to enhance learning by aligning methods with user profiles [42], concerns about their theoretical validity persist [55, 56]. Critics argue that learning styles lack scientific backing and may pigeonhole learners, limiting their educational development.

In contrast, with arguably competing ideological implications [15], the work of Benjamin Bloom has been extremely influential for education sciences and adaptive learning systems. Bloom's Taxonomy, introduced in 1956 [45], is a foundational framework categorizing cognitive skills into six hierarchical levels of cognitive complexity: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating [6]. It has significantly influenced intelligent tutoring systems, from informing the structure of knowledge models by mapping learning objectives with the different cognitive complexity levels [11] to providing a framework for the evaluation of e-learning applications [35]. Building on this taxonomy, Bloom developed Mastery Learning, which uses a hierarchical structure to guide learning and instruction [34]. Mastery learning emphasizes clear instructional objectives, sequential learning units, regular diagnostic assessments, and time flexibility—all aimed at ensuring each student reaches mastery for each learning unit before progressing, ensuring deeper learning and more equitable outcomes among students [15].

2.2 AI and Adaptive Learning Systems

Over the years, adaptive educational systems have evolved from basic rule-based methods to highly sophisticated AI-driven systems [60]. Machine learning (ML) techniques, have been used to infer and construct student profiles [19], to model knowledge levels and perform knowledge tracing [20], to generate and adapt content dynamically [29], and recommend study material [31]. Natural language processing and generation techniques has been used for automatic question generation [53], automatic assessment [22], quiz generation [47], and question difficulty estimation [13]. Other applications, especially with recent developments in the field of LLMs, include natural-language based systems such as chatbots [21] and virtual teaching assistants [12], that offer an increasingly personalized experience to students. Question generation from text has been further developed by incorporating Retrieval-Augmented Generation (RAG), which harnesses the power of LLMs to generate questions grounded in specific learning materials. This approach helps avoid hallucinations and ensures that the generated questions are contextually relevant and accurate [28, 36].

2.3 Similar Systems

2.3.1 Research. Adaptive learning systems have been shown to enhance learning outcomes and user satisfaction. Systems that adapt based on the user’s knowledge level and goals have led to higher exam scores and positive positive [43]. Similarly, personalizing content based on learners’ knowledge levels helps achieve goals more efficiently, resulting in higher post-test scores, fewer course steps, and increased satisfaction [68]. Models that adapt course content using both initial student knowledge and real-time learning analytics also lead to higher performance and efficiency [9].

It has also been shown that mastery-based adaptive techniques, such as performance-based flashcard dropping, improve long-term knowledge retention [67]. Modeling forgetting curves for spaced repetition in flashcard systems also improves memory retention [69], and considering time as a contextual factor can inform adaptation by aligning materials with learners’ pacing [63].

Bloom’s Taxonomy has frequently been used in adaptive learning systems to enhance learning outcomes. Grubišić et al. [32] designed and evaluated a system that automatically generates course materials using Bloom’s taxonomy, Bayesian networks, and learner stereotypes (Novice to Expert). Additionally, Bloom’s Taxonomy has been applied to adapt test item selection and difficulty [11, 18].

Finally, tutoring systems that use advanced NLP technology and LLMs can easily serve as “general-use” tools offering a great range of domain-independent features such as automatic course planning, material generation, and adaptive quizzes, almost substituting tutors [17, 21].

2.3.2 Market. Anki [5] is a system for manual flashcard creation, learning and repeated self assessment. Its algorithm using spaced repetition optimizes content memorization [41]. However, the focus seems to be primarily on recall rather than higher cognitive levels. Khan Academy [3] is an online learning platform which integrates principles of mastery learning with quizzes, mastery exams, and elements of gamification, alongside personalization options such as goal selection. Even though it is a course-based platform, it showcases the effectiveness of this approach for both pupils and tutors [57, 65].

Revisely [1] provides automated flashcards and quizzes, but it does not use a pedagogy-informed framework for knowledge modeling. Quizlet [2] offers a sophisticated algorithm for flashcard presentation and quizzes creation with varying question difficulty, but lacks the AI-generated flashcards feature. MemoryLab [4] has an emphasis on the research-driven algorithm for flashcard sequencing and revision, but does not seem to incorporate varying question complexity.

Finally, teams at Google are developing a conversational AI tutor called LearnLM-tutor. This LLM-based system is designed to provide personalized educational support through a participatory and evaluation-driven approach, fine-tuned on pedagogical principles [39, 40], and could significantly disrupt the Educational Technology (EdTech) industry.

At the same time, Google’s efforts highlight that LLM-powered intelligent tutoring systems are more relevant than ever. In this light, our proposed solution aims to integrate all the missing features of current commercial systems by incorporating automated quiz creation, adaptive question difficulty, mastery-based knowledge and student modeling, and a flashcard-style learning interface.

3 SYSTEM AND MODEL

The adaptive educational system proposed here is designed to personalize the learning process based on students’ own material, specific goals, and preferences, dynamically adapting to their learning progress. This is mainly achieved through: 1) a mechanism for gradually recommending questions of higher complexity and specificity (based on Bloom’s Taxonomy) and 2) adjusting the presented material based on the student’s performance on these questions. Both the pace of learning progression and the intended mastery level are specific to the individual student and the material they are currently studying.

For a comprehensive overview of the terminology used in this section, please refer to Appendix Section A.

3.1 User Flow and Interactions

3.1.1 Basic User Flow. Following Mastery Learning theory, the student learns in Modules as in [59]. The student creates a Module and can upload their own material in PDF format (for simplicity we will assume one large PDF document, such as a textbook). After creation of a Module, the student is prompted to complete a Calibration Phase where the student sets learning preferences (learning goal and preferred pace). This is done to address the new user cold start problem, which occurs when there is little to no information available about a user, making it difficult to generate an initial profile [30]. See Section 3.1.2 for a detailed presentation of the Calibration Phase.

The user can navigate into a specific Module and add material, see their overall progress, or begin a Learning Session. A Learning Session consists of iterative Learning Cycles. A Learning Cycle consists of a sequence of Cards presented in succession and a Quiz. Each Card corresponds directly to an excerpt of the PDF and contains the respective text. During the succession, the user can select “Next” or “Previous” to navigate between the Cards or select “Ignore this Card” to exclude this excerpt from the Module. This means that the respective excerpt will be excluded from the Cards set and will not appear in future Learning Sessions or Learning Cycles. This is done so that the student has some control over the learning process and dynamically

adapt the material that most closely aligns with her goals as the user explores it through the system (see Figure 5 for an example of a Card).

When at the end of the Card sequence, the student has to complete a Quiz related to the previously presented Cards. The Quiz contains auto-generated Multiple Choice Questions (from now on referred to as Questions) of different difficulties. The difficulties correspond to Bloom’s Taxonomy as described in Section 3.2.3. After submitting the Quiz, the student sees the score, the correct answer, and the explanation of the answer (Fig. 5). Then a new Learning Cycle begins with an updated Card set and a new Quiz. The whole iterative process can be seen in Figure 1. The student can exit the Learning Session and go back to the Module screen or choose to learn another Module. The system keeps updates up to the latest submitted Quiz.

When the Target Mastery Level is reached for all valid (non-ignored) Cards the student is recommended by the system to take the Module Exam. As in Khan Academy and according to Mastery Learning theory [15] learners retain knowledge more effectively when assessments and learning content are broken down into smaller, manageable parts. The Module Exam can also be taken at any time the student feels ready.

3.1.2 Calibration Phase. Before starting a Learning Session for a newly created Module, the student is prompted to go through an initial Calibration Phase where they can specify their learning objectives (mastery goals) for the specific Module, select their preferred learning pace, and assess their previous knowledge on the material in the Module. See Appendix Section B for the whole questionnaire.

Learning Goals Setting: Through a set of questions corresponding to the 6 Levels of Bloom’s Taxonomy, students set their desired learning goals for the Module. This determines the maximum Mastery Level that is reached for this Module (Target Mastery Level, see how it affects the knowledge model in Section 3.2.2).

Pace Preference Specification: Students specify their preferred learning pace (slow, medium, fast), which determines the Mastery Progression Rate Parameter α (see how it affects the knowledge update in Section 3.2.4).

Previous Knowledge Assessment: The student takes a Module Exam to assess their current understanding of the Module content. This is essentially an extended Quiz containing Questions that cover all Cards and are of various difficulties. This way an initial Mastery Score is assigned to each Card, based on the student’s pre-existing knowledge. Assessing previous knowledge is a way of personalization important in adaptive learning systems [16]¹.

3.1.3 Progress Visualisation. Based on Mastery Learning theory, progress visualization is crucial for enabling students to monitor their understanding and identify areas needing improvement. For this reason, we propose a way for the student to visualise their progress on the Module by inspecting the PDF. Each excerpt (corresponding to a Card) is colored either in Red (representing a low Mastery Score, lower than 0.2), Yellow (score from 0.2 to 0.8), and Green (score above 0.8) (refer to Figure 4 for the wireframe and Section 3.2.1 for the explanation of Mastery Scores). Excerpts that are not yet seen are

in Grey and ignored excerpts are hidden with an indication in the place where they would normally be. The color palette follows conventions for performance that may not be universal across cultures (see Limitations).

3.2 Modeling Student, Item, and Knowledge

3.2.1 Student Model. The student model captures the learner’s progression and interaction with the system. Each excerpt of the material (Card) is associated with a Mastery Score, a continuous value ranging from 0 to 1, representing the student’s understanding of that specific piece of content. These scores are dynamically updated based on Quiz performance. Mastery Scores are further categorized into 5 Learner Stereotypes as seen below (Novice, Beginner, Intermediate, Advanced, and Expert), which guide the adaptive system in selecting appropriate questions for the student. The mapping follows [32], who used learner stereotypes to recommend courseware material. In addition, the student model tracks learning preferences (mastery goals and learning pace) and interaction history, including Cards ignored and Quiz results.

The continuous Mastery Score for each Card $M(c)$ is mapped to a discrete level $L(M(c))$, corresponding to Learner Stereotypes, as follows:

$$L(M(c)) = \begin{cases} 0, & \text{if } M(c) < 0.2 \quad (\text{Novice}) \\ 1, & \text{if } 0.2 \leq M(c) < 0.4 \quad (\text{Beginner}) \\ 2, & \text{if } 0.4 \leq M(c) < 0.6 \quad (\text{Intermediate}) \\ 3, & \text{if } 0.6 \leq M(c) < 0.8 \quad (\text{Advanced}) \\ 4, & \text{if } M(c) \geq 0.8 \quad (\text{Expert}) \end{cases}$$

3.2.2 Knowledge Model. The knowledge model represents the structure of the learning material. Each Module consists of a collection of Cards corresponding to excerpts of the document(s). The Mastery Score for each Card reflects the student’s proficiency on that piece of the material. Mastery Scores are categorized into the 5 Learner Stereotypes as mentioned above (Section 3.2.1). A Card is considered “learned” (boolean) if its Mastery Level (corresponds to learner stereotypes) reaches the Target Mastery Level (the learner stereotype set by the student in the Calibration Phase). This also means that, for example, if the student sets the highest Mastery Level to Advanced (by answering the questionnaire), no Question will have a higher difficulty than that (refer to Section 3.2.3 for the explanation of questions difficulty). The system determines Module Mastery by evaluating whether the learner’s Mastery Score for each valid (non-ignored) Card is learned. Due to the dynamic nature of Mastery Score, it can drop again below 0.8, which renders the Card “not learned”.

3.2.3 Item Model. The Item Model governs the generation and classification of Questions used to assess the student’s knowledge of the material. Multiple choice questions (including correct answers and explanation of the answers) are generated from text and classified into 6 levels of complexity (or difficulty) according to Bloom’s Taxonomy following a RAG and classification pipeline as described in [33]. In our system, levels 5 and 6 are both mapped to the Expert learner stereotype to facilitate integration with the rest of the system. Building upon [33], each question is linked to one or more Cards using a cosine similarity function to measure relevance. The cosine similarity between the Question q and each Card c is computed as:

¹The exact algorithm of the Module Exam in the Calibration Phase is not described in the current study. It should be efficient so it is quick for the user and a Mastery Score is determined for all or most of the material.

$$\text{cosine_similarity}(q, c) = \frac{\mathbf{q} \cdot \mathbf{c}}{\|\mathbf{q}\| \|\mathbf{c}\|} \quad (1)$$

A softmax weighting function determines the contribution weight $C_{c,q}$ of each of the top- k Cards to a given question:

$$C_{c,q} = \frac{\exp(\text{cosine_similarity}(q, c))}{\sum_{c' \in \text{top-}k} \exp(\text{cosine_similarity}(q, c'))} \quad (2)$$

The system dynamically generates Quizzes by selecting Questions appropriate to the specific student's progress for each Card (see Section 3.2.5).

3.2.4 Knowledge Update. After the student submits a Quiz, the system updates the Mastery Scores of the associated Cards using the Mastery Update Rule:

$$M(c) = M(c) + \alpha \times C_{c,q} \times \delta \quad (3)$$

where $\delta = 1$ if the answer is correct, and $\delta = -1$ if incorrect. Correct answers increase the Mastery Scores $M(c)$, while incorrect answers decrease them. The adjustment magnitude depends on the contribution weight $C_{c,q}$ and the Mastery Progression Rate Parameter α as seen in Equation 3.

The pace preference set by the student during the Calibration Phase determines the value of the Mastery Progression Rate Parameter (α). The exact value of α can be refined through simulation and user testing, and future research may suggest that this parameter should be dynamic. In any case, the ability of the student to set an initial value to this parameter allows them to tailor the learning experience to their preferred pace and engagement level.

For a visualization of how Questions are linked to Cards and how the system follows rule-based adaptation for the creation of a new Learning Cycle, refer to Figures 2 and 3.

3.2.5 Adaptation Rules. After each Quiz, the system adapts the learning material based on the student's performance, using the following rules:

Rule 1: Card Selection. Let C be the set of all Cards, and $I \subseteq C$ be the set of ignored Cards. The set of valid Cards is $V = C \setminus I$. Each Card $c \in V$ has an associated Mastery Score $M(c)$.

The probability $P(c)$ of selecting a Card c for the next Learning Cycle is inversely proportional to its Mastery Score and adjusted with a prior probability to ensure that new or unseen Cards are not neglected. This can be expressed as:

$$P(c) = \frac{(1 - M(c)) + \gamma}{Z} \quad (4)$$

where γ is a small positive constant representing the prior probability, and

$$Z = \sum_{c \in V} ((1 - M(c)) + \gamma)$$

is a normalization constant ensuring that the probabilities sum to 1.

This design effectively promotes a uniform progression of Mastery Levels across all cards. By focusing practice on the weaker cards, the system aims to elevate their Mastery Scores, bringing them closer to the levels of cards the student has already mastered.

If the student's total Quiz score in the current Learning Cycle exceeds a predefined threshold θ , the system increases the prior

probability γ for unseen Cards (Cards with $M(c) = 0$) to encourage the introduction of new material:

$$\gamma_{\text{new}} = \gamma \times \delta \quad (5)$$

where $\delta > 1$ is the boosting factor. The exact threshold θ can be determined after simulation and user studies and can also be adaptive.

Rules 2 and 3: Question Selection. Questions are filtered based on the student's Target Mastery Level T set during the Calibration Phase. Only Questions with a difficulty level $S(q)$ less than or equal to T are considered. Each Question q is linked to one or more Cards using a cosine similarity function, and only Questions linked to the Cards selected for the current Learning Cycle are included in the Quiz.

Rule 4: Module Exam Recommendation. The system recommends that the student takes the Module Exam when the Mastery Score $M(c)$ for all valid Cards $c \in V$ meets or exceeds the Target Mastery Level T :

$$\forall c \in V, \quad M(c) \geq T \quad (6)$$

3.2.6 Time-Based Decay in Mastery Scores. According to Ebbinghaus's Forgetting Curve, human memory retention diminishes rapidly over time, with the majority of forgetting taking place soon after initial learning—between 20 minutes and 24 hours [54]. Inspired by studies mentioned above that factor in time as a contextual element and guided by the principles of spaced repetition, Mastery Scores in our system decrease over time to reflect natural memory decay. By doing so, the system supports periodic review of the material, aiding in the consolidation of information.

The decay is governed by the decay multiplier (λ) and the Mastery Decay Parameter (β). The Mastery Score ($M(c)$) decreases according to the elapsed time (τ) since the student's last interaction with the specific Module.

The decay function is defined as:

$$M(c) = M(c) \times e^{-\beta\lambda\tau} \quad (7)$$

where:

- $M(c)$ is the updated Mastery Level.
- β is the Mastery Decay Parameter.
- λ is the decay multiplier based on context.
- τ is the time elapsed since the student's last interaction.

The decay multiplier (λ) adjusts the rate of decay based on the context:

$$\lambda = \begin{cases} 0, & \text{if } \tau < 4 \text{ hours} \\ 1, & \text{if } \tau \geq 4 \text{ hours and same date} \\ 2, & \text{if } \tau \geq 4 \text{ hours and different date} \end{cases} \quad (8)$$

It should be noted that this approach serves as a template for context-based adaptation in our system. The parameters β and λ can be adjusted after user studies or made dynamic to better align with individual learning patterns.

4 EVALUATION

To ensure the proposed system is successful in filling the aforementioned gaps, refining the parameters discussed in Section 3.2, all with a high level of user satisfaction, its key components as well as the system as a whole will be evaluated. This includes testing the effectiveness and satisfaction level of the flashcard system, testing

the quality of the content on the flashcards produced by the LLM, validating the adaptation rules and do a competitor analysis.

For the evaluation of the RAG model, a different approach is needed than for traditional machine learning algorithms, as there is not discrete output. The RAG generates text of variable length, where two very similar looking sentences can have a complete different meaning. There are different metrics available for evaluating RAG models like BLEU, METEOR, NIST and more, but in this case ROUGE (Recall-Oriented Understudy for Gisting Evaluation) [49] is chosen as the evaluation tool, as ROUGE is a widely used evaluation metric for assessing the quality of automatic summaries generated by text summarization systems. The generated summary is compared with a reference, ie. human generated, summary. This is a useful method as the flashcards are basically a very short summary of the provided learning material. ROUGE focuses on the recall score, which measures how much of the important information from the reference summaries is captured by the generated summary. Higher ROUGE scores indicate better performance in terms of preserving key information from the original text, which is a metric of determining the quality of the produced question. A qualitative quality measure will be provided by the teaching professionals that are tasked to provide the human generated questions. The generation of the human questions and the ROUGE evaluation will be conducted in the controlled environment of a testlab.

The modelling process, especially the item modelling, is deemed too complex for user testing. As such, the prediction algorithms are reviewed by experts. These experts should have extensive knowledge in the field of adaptive systems and will check our system against current industry standards to insure the proper implementation of our card selection, mastery levels, decay function and the dynamic adjustment of our system parameters. After the proper function is verified, an uncontrolled A/B test will be conducted. One system will have the algorithm for flashcard selection and the other will not. Both system will use the same pre-defined learning materials. This evaluation will be done in order to gather and compare important performance metrics such as user retention and time to reach the learning goal for example. The system data will be logged and analyzed afterwards. In order to qualitatively explain the data, user are asked to fill in the e-learning satisfaction scale (eLSS) [38]. The eLSS is a valid and reliable scale for quickly and easily measuring experiences with distance learning and the effectiveness of e-learning. The eLSS consists of 17 items with three factors: content, interface, and communication.

As a system-as-a-whole evaluation, the system is tested against Anki, the flashcard based learning system that was discussed in ???. This popular study tool uses the spaced repetition approach, which involves reviewing information at increasing intervals. The evaluation process consists of a longitudinal user study. The participants are split into two groups, either using Anki or the proposed system for a duration of two weeks, during which they are asked to keep a diary in which they reflect on their user experience. The metrics for this evaluation methods are ease of use, effectiveness and user satisfaction. To get consistent results the user will be handed by a list of pre-defined questions to guide the diary process.

5 DISCUSSION

Based on the feedback we received from our peers, we decided to revise several design decisions for the system. First, we modified the mastery score model and addressed concerns about the difficulty level of the questions. Second, we improved the interface design to offer more detailed feedback after each assessment phase. Instead of displaying short numerical messages as results, we now explicitly show users which questions they answered incorrectly and allow them to review the correct answers. Lastly, we reintegrated some of the mathematical formulas into the main content and provided thorough explanations for better understanding.

Some other feedback we received was not incorporated in the model so we reflect on those as limitations of the project.

5.1 Limitations

5.1.1 Limitations of the System. Like all software products, this educational system has several limitations that need to be taken into account. Some of these limitations present opportunities for extending and improving the system. Given the already crowded market of educational technology, it is crucial to understand both the unique factors and limitations of the system.

Firstly, the system cannot handle study materials that include multimedia elements such as images or videos, limiting its application to text-based content. Furthermore, the system heavily relies on the user's motivation, making it less effective for learners who are not self-driven. Since the results are compared to the user's personal learning goals, the system cannot predict exam outcomes or guarantee that the user's knowledge is sufficient.

Additionally, the AI performance within the platform can be inconsistent, affecting the quality of content segmentation and assessment generation. The system may also be less suitable for learners with different learning styles, particularly those who benefit from more interactive or hands-on methods. Lastly, the material segmentation process has not been fully developed or rigorously tested, as designing the user experience and pre-processing steps for new materials was beyond the scope of this project.

5.1.2 Limitations of the Evaluation. Some of the known limitations of the ROUGE evaluation method include: (1) Dependence on human-created reference summaries. Not only can the quality and accuracy of the reference summaries affect the ROUGE-score, it can also introduce subjectivity and variability when different researchers use different sets of reference summaries. The ROUGE-score is also primarily focused on recall, but different researchers might prioritize other factors like precision or fluency. (2) Lack of contextual awareness. The broader context in which a summary is generated is not taken into consideration and (3) Capture semantic similarity. Lexical overlap between summaries may not always be sufficient for capturing more complex aspects of language, like paraphrasing or semantic similarity.

5.2 Ethics

The functioning of the system as described currently, with the uploading by users of material raises some concerns about copyright infringement. A primary concern would be the misuse of the uploaded material, as some LLMs could store, reproduce or redistribute the general content of the uploaded material through their learning

mechanisms. Another issue as well for the platform is the possibility that the copyright prohibits the use of the material in a LLM, which the platform might be held responsible for. Additionally the material could be misinterpreted by the LLM, which could lead to reputational damage for the author.

5.3 Options to Extend the System

There are several ways in which the system could be improved from the design presented in this paper. As mentioned previously in the limitations, the system currently does not accept input in any other form than written. If the system were able to interpret video and/or image content as well, this could help users practice a wider variety of learning material. The addition of gamification features as well could help users maintain their learning momentum, make them stick to doing the work required to complete the learning challenge they set themselves. Finally we would like to explicitly work out various ways in which the system could be made more accessible to users with disabilities. This could for example be features such as voice assistants to help navigate the UI or the implementation of alternate colour schemes.

6 CONCLUSION

The adaptive educational system was designed to address key challenges in individual learning, particularly the segmentation of study material, the lack of contextual support in written resources, and the subjectivity of self-assessment. This system provides a solution for young adults to study in an iterative manner, with each cycle consisting of a learning phase followed by a self-assessment phase. It guides users through the process with a streamlined interface and adapts the study material based on an evolving user model.

What sets the system apart is its content-agnostic nature and the use of AI in multiple steps. AI is employed for content segmentation, context provision, and during the assessment phase, where it generates customized questions tailored to the user's specific goals. Additionally, the system is highly scaleable and easy to adopt, as it is delivered as a web application with a simple single login, making it accessible to users across various platforms without the need for complex installations or technical setups.

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APPENDIX A TERMINOLOGY

- **Calibration Phase:** An initial phase where the student sets learning preferences, such as mastery goals and preferred learning pace, and assesses prior knowledge.
- **Card:** An excerpt of the learning material (e.g., from a PDF) presented to the student.
- **Learner Stereotypes:** Categories used to classify students based on their Mastery Scores (e.g., Novice to Expert).
- **Learning Cycle:** Consists of a sequence of Cards followed by a Quiz.
- **Learning Session:** A session during which the student engages with the learning material through Learning Cycles.
- **Mastery Decay:** The decrease in Mastery Scores over time when the material is not revisited for a certain time.
- **Mastery Decay Parameter (β):** A parameter controlling the rate at which Mastery Scores decrease over time due to simulated memory decay (to encourage revision).
- **Mastery Level:** A categorical level of proficiency, such as Novice, Beginner, Intermediate, Advanced, or Expert (directly corresponding to Learner Stereotypes, see below).
- **Mastery Progression Rate Parameter (α):** A parameter determining the rate at which Mastery Scores are updated, determined by the student's preferred learning pace.
- **Mastery Score:** A continuous value between 0 and 1 representing the student's understanding of a specific Card.
- **Mastery Update Rule:** The algorithm used to update Mastery Scores based on Quiz performance.
- **Module:** A collection of learning material centered around a specific topic or subject area.
- **Questions:** Auto-generated multiple-choice questions of varying difficulties, aligned with Bloom's Taxonomy.
- **Quiz:** A set of Questions assessing the student's understanding of the material covered in the sequence of Cards.
- **Target Mastery Level:** The desired Mastery Level set by the student during the Calibration Phase.

APPENDIX B INITIAL CALIBRATION QUESTIONNAIRE

1. Set Your Learning Goal

Please select the statement that best describes your learning objective for this Module:

- A. I want to **memorize** key facts and basic concepts.
- B. I want to **comprehend** the material and explain ideas in my own words.
- C. I want to **use** the information to solve problems or perform tasks.
- D. I want to **examine** relationships and **break down** concepts into parts.
- E. I want to **critically assess** concepts and make judgments based on criteria.
- F. I want to **develop** new ideas or products using the concepts learned.

2. Set Your Learning Pace Preference

Please select the statement that best describes your learning pace preference for this Module:

- A. I enjoy making rapid progress and don't mind quickly moving through material, even if it means the challenge is greater.
- B. I prefer a steady pace, balancing speed with thorough understanding.
- C. I like to take my time, carefully studying each piece of information to fully grasp the details and testing myself often, even if it means progressing more slowly.

3. Please Take the Module Exam

In the following exam you will be tested on your existing knowledge on the Module material. This process can take long. You can stop at any time. The system will keep the answers given up to that point.

APPENDIX C SYSTEM AND MODELING FIGURES

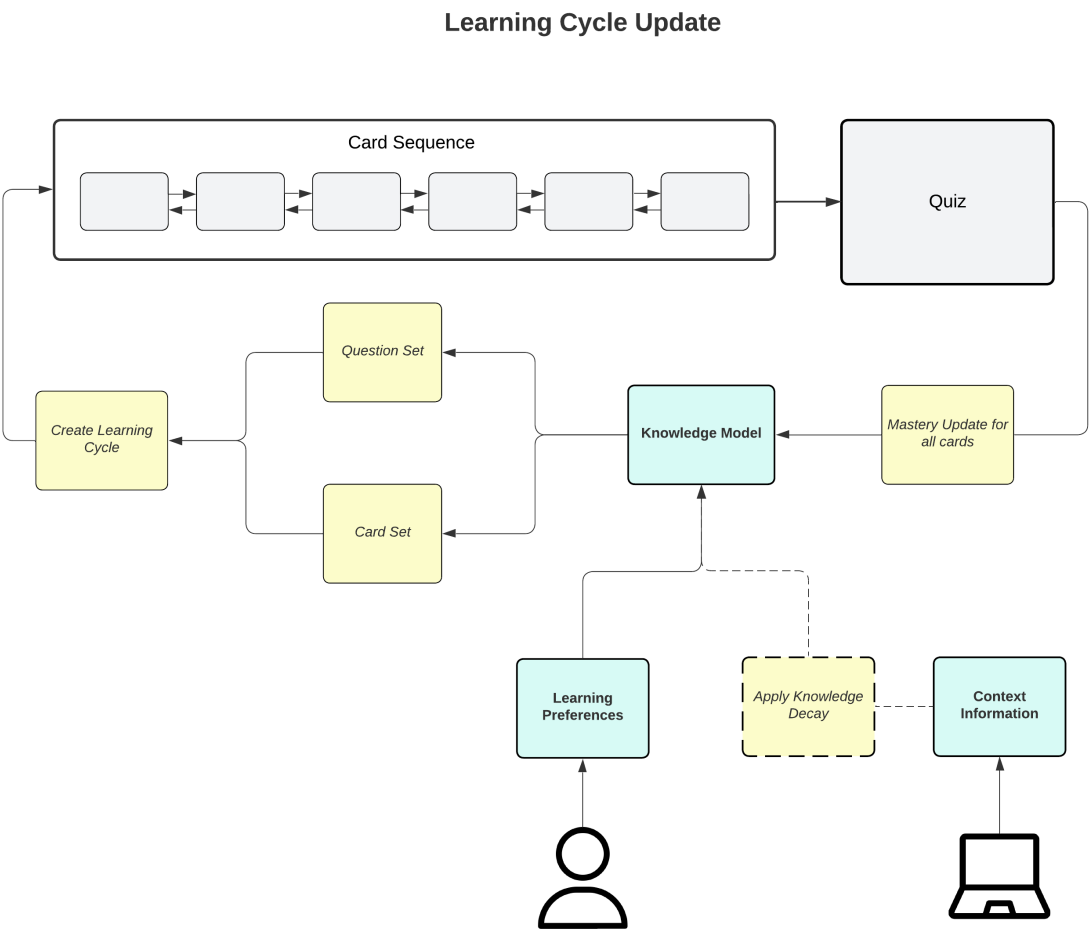


Figure 1: Learning Cycle Update

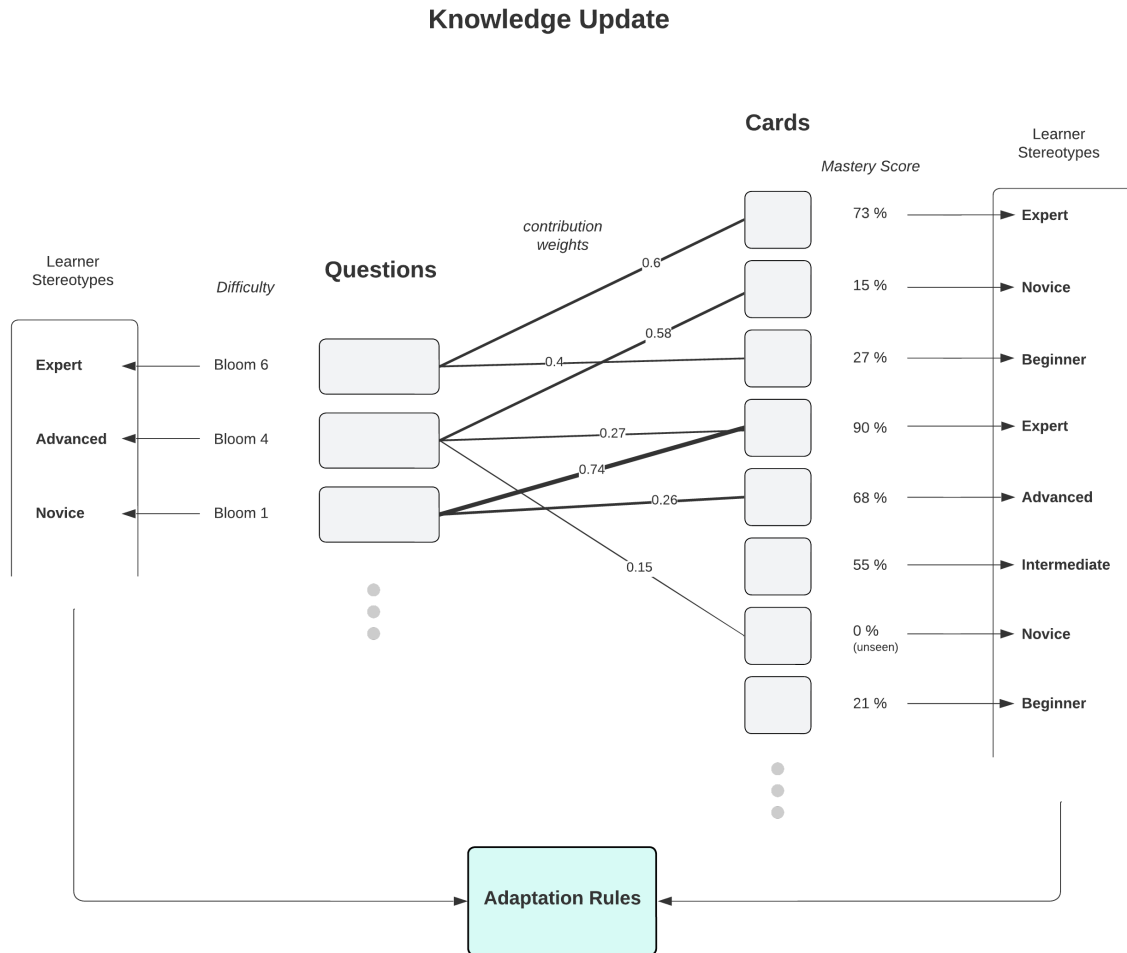


Figure 2: Knowledge Update

Adaptation Rules

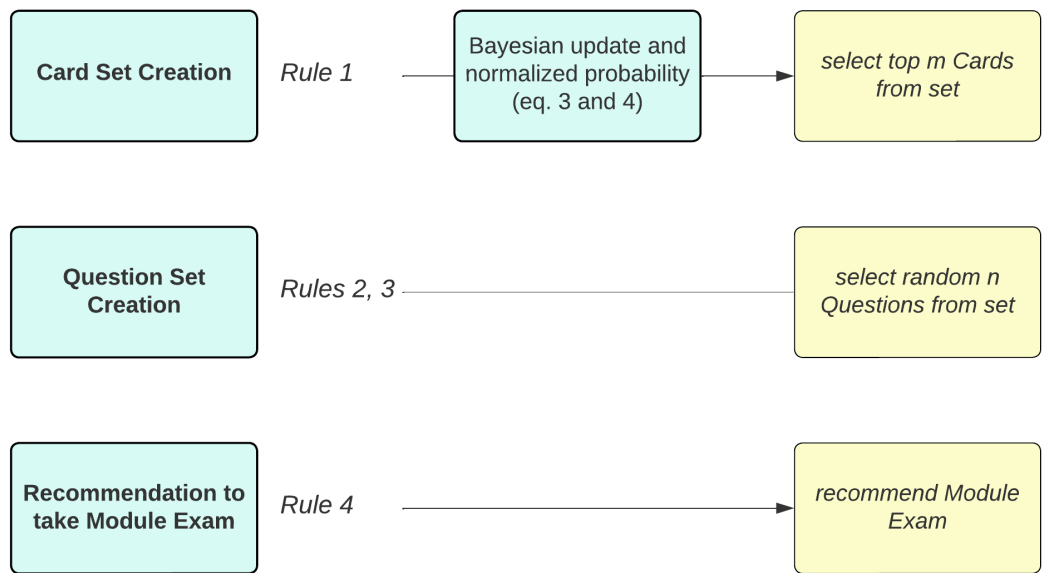


Figure 3: Adaptation Rules

APPENDIX D INTERFACE DESIGN

1

Navigation and revision

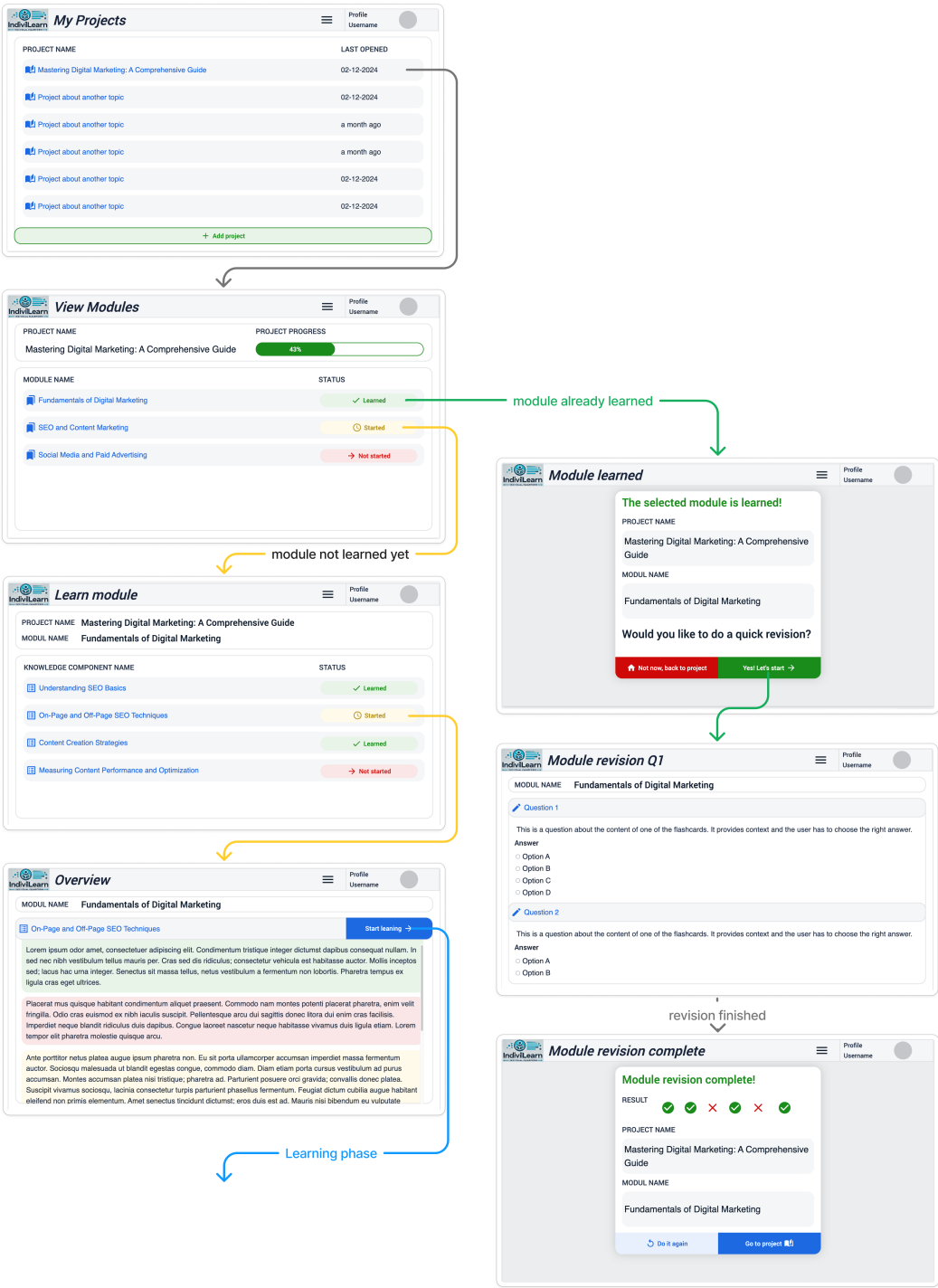


Figure 4: User interface and interaction flow from the main screen until entering the learning and assessment phase

Learning phase

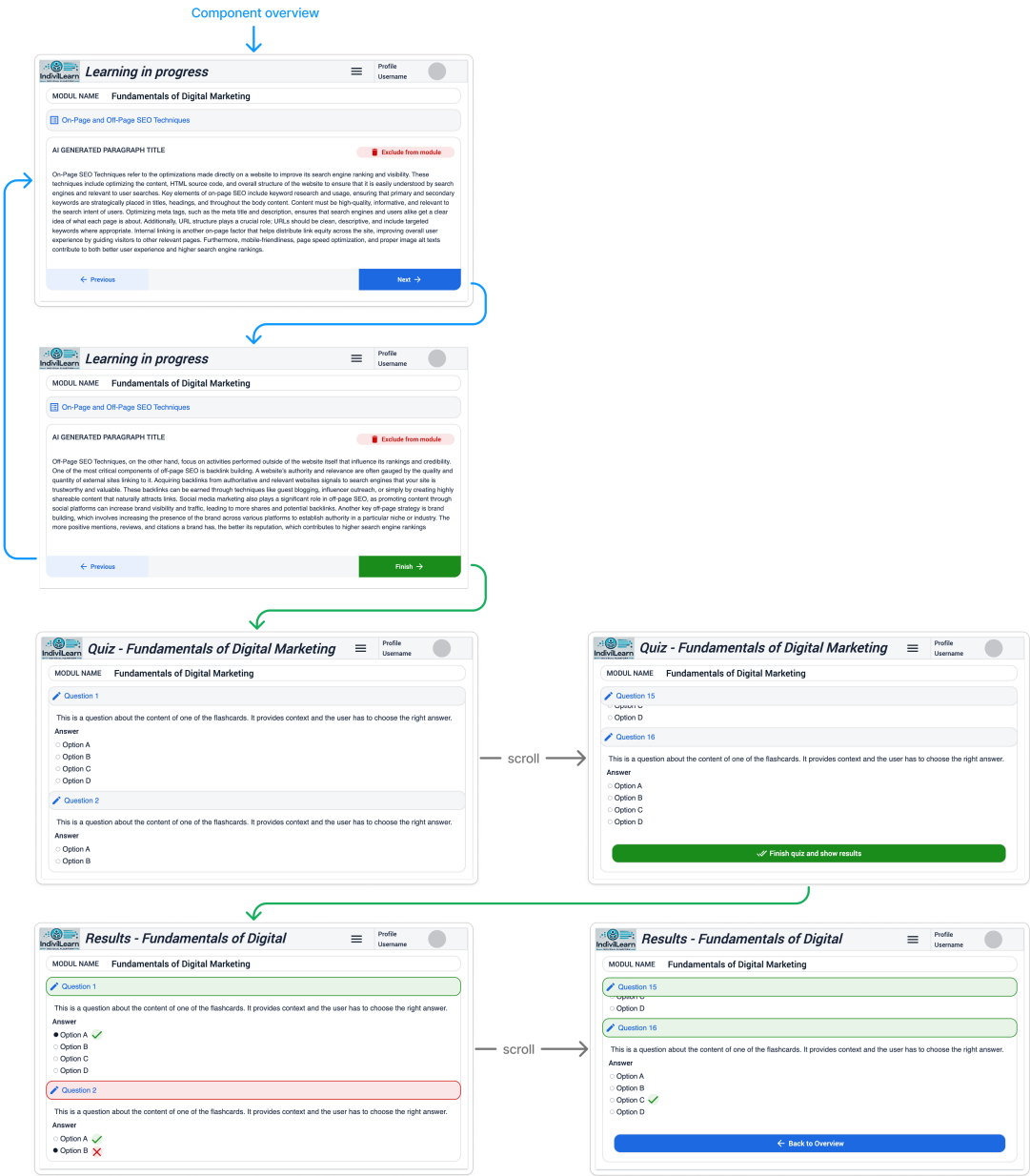


Figure 5: User interface and interaction flow of the iterative learning phase and quiz